

SANDAKAN, MalaysiaWMO: **964910**Lat: **5.901N**Lon: **118.059E**Elev: **14**StdP: **101.16**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.9	23.3	21.5	16.1	26.9	21.9	16.6	26.5	8.1	27.3	7.3	27.5	1.9	260	0.349

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.1	33.9	26.7	33.2	26.6	32.8	26.6	27.8	31.5	27.6	31.2	27.2	30.8	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	30.4	26.4	21.9	29.7	26.2	21.6	29.5	89.4	31.4	87.5	31.7	86.4	31.4	34.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.4	5.0	21.7	35.7	1.3	1.6	20.7	36.8	19.9	37.8	19.2	38.7	18.2	39.8		
(5)			20.9	29.1	1.2	1.3	20.1	30.0	19.4	30.8	18.7	31.6	17.9	32.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	26.9	27.1	27.6	28.2	28.7	28.5	28.3	28.4	28.3	27.9	27.6	27.3	
	DBStd	1.08	1.07	1.03	0.89	0.81	0.86	0.92	0.97	0.94	0.89	0.85	0.81	0.99	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6539	524	478	547	547	581	554	568	572	548	556	528	537	
	CDD18.3	3497	266	245	288	297	322	304	309	313	298	297	278	278	
	CDH23.3	37569	2554	2453	3105	3468	3791	3406	3367	3407	3193	3149	2872	2806	
	CDH26.7	13152	604	616	982	1320	1554	1350	1315	1362	1242	1118	903	787	
Wind	WSAvg	2.2	2.8	3.0	2.7	2.2	2.0	1.8	1.8	1.9	1.8	1.8	1.9	2.3	
Precipitation	PrecAvg	3224	506	292	203	139	150	189	200	253	244	227	338	483	
	PrecMax	4130	1256	712	531	348	348	338	437	477	519	356	815	884	
	PrecMin	2515	112	48	45	14	53	56	70	85	86	92	174	148	
	PrecStd	537	318	188	139	103	85	80	101	102	127	79	143	221	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.2	32.0	33.2	34.2	34.1	34.2	34.2	34.1	33.6	32.8	32.1	
		MCWB	26.4	26.2	26.6	27.1	27.2	26.8	26.6	26.4	26.4	26.6	26.3	26.5	
	2%	DB	30.2	30.3	31.2	32.2	33.1	33.2	33.2	33.5	33.1	32.2	31.6	31.1	
		MCWB	26.2	26.0	26.4	26.8	27.1	26.7	26.4	26.3	26.4	26.4	26.4	26.7	
	5%	DB	29.9	30.0	30.9	31.9	32.4	32.4	32.4	32.8	32.2	31.6	31.0	30.5	
		MCWB	26.1	25.9	26.3	26.7	26.9	26.6	26.4	26.2	26.4	26.5	26.6	26.5	
	10%	DB	29.2	29.6	30.2	31.2	31.9	31.9	31.9	31.7	31.9	31.6	31.0	30.2	29.9
		MCWB	25.7	25.7	26.0	26.5	26.9	26.6	26.4	26.2	26.4	26.4	26.3	26.3	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.1	27.3	27.9	28.3	28.1	28.1	27.8	27.7	27.6	27.5	27.6	
		MCDB	29.7	29.8	30.5	31.7	32.3	31.8	31.4	31.5	31.6	31.0	30.6	30.4	
	2%	WB	26.7	26.6	26.9	27.4	27.7	27.6	27.4	27.2	27.2	27.1	27.1	27.1	
		MCDB	29.2	29.4	30.2	31.1	31.6	31.4	31.1	31.2	31.1	30.8	30.2	29.8	
	5%	WB	26.3	26.2	26.6	27.1	27.4	27.2	27.0	26.8	26.9	26.8	26.8	26.8	
		MCDB	28.7	28.9	29.7	30.6	31.2	31.0	30.9	30.9	30.8	30.3	29.8	29.3	
	10%	WB	26.1	26.0	26.2	26.7	27.1	26.8	26.6	26.5	26.6	26.5	26.5	26.5	
		MCDB	28.4	28.6	29.2	30.1	30.7	30.5	30.4	30.4	30.3	29.9	29.4	28.9	
Mean Daily Temperature Range	5% DB	MDBR	4.7	4.9	5.6	6.4	7.1	7.4	7.5	7.9	7.7	6.9	6.0	5.3	
		MCDBR	5.3	5.3	6.0	6.9	7.7	8.2	8.4	8.8	8.5	7.7	6.8	6.0	
		MCWBR	2.5	2.4	2.5	2.6	2.9	3.2	3.3	3.4	3.4	3.1	2.9	2.6	
	5% WB	MCDBR	5.0	5.1	5.8	6.5	7.3	7.8	7.9	8.2	8.0	7.1	6.3	5.6	
		MCWBR	2.6	2.5	2.6	2.8	3.1	3.3	3.4	3.4	3.5	3.1	2.9	2.7	
Clear-Sky Solar Irradiance	taub	0.425	0.424	0.419	0.420	0.406	0.406	0.410	0.436	0.446	0.453	0.414	0.414		
	taud	2.462	2.441	2.453	2.442	2.506	2.513	2.501	2.403	2.369	2.347	2.483	2.497		
	Ebn at Noon	895	903	905	889	882	871	871	862	867	864	898	898		
	Edh at Noon	114	119	118	118	107	105	107	121	127	129	111	109		
All-Sky Solar Radiation	RadAvg	4.22	4.86	5.38	5.68	5.56	5.18	5.20	5.37	5.31	4.96	4.54	4.29		
	RadStd	0.43	0.47	0.45	0.35	0.33	0.26	0.28	0.27	0.33	0.25	0.31	0.41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.30	+0.38	+0.29	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.13	N/A	N/A	+0.27	+0.34	+0.25	N/A	N/A	+94	+94

Nomenclature: See separate page